

81. On heating pyrolusite with KOH in presence of air we get

- (a) KMnO_4
- (b) K_2MnO_4
- (c) $\text{Mn}(\text{OH})_2$
- (d) Mn_3O_4

Answer: B — (b) K_2MnO_4

Chemical reason: Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result K_2MnO_4 is consistent with that behaviour.

82. $\text{Cu}(\text{CN})^{42-}$ is colourless as it absorbs light in

- (a) Visible region
- (b) Ultraviolet region
- (c) Infrared region
- (d) All above are wrong

Answer: D — (d) All above are wrong

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in All above are wrong has the characteristic oxidation state/ligand field that produces the observed colour.

83. Acidified solution of chromic acid on treatment with hydrogen peroxide yields

- (a) $\text{CrO}_3 + \text{H}_2\text{O} + \text{O}_2$
- (b) $\text{Cr}_2\text{O}_3 + \text{H}_2\text{O} + \text{O}_2$
- (c) $\text{CrO}_5 + \text{H}_2\text{O}$
- (d) $\text{H}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O} + \text{O}_2$

Answer: C — (c) $\text{CrO}_5 + \text{H}_2\text{O}$

Chemical reason: The correct choice is $\text{CrO}_5 + \text{H}_2\text{O}$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

84. Which of the following metals corrodes readily in moist air

- (a) Gold
- (b) Silver
- (c) Nickel
- (d) Iron

Answer: D — (d) Iron

Chemical reason: Iron corrodes readily in moist air because water and oxygen support electrochemical rusting. Gold is noble, while Ni is protected by a passive surface film.



85. Which one of the following compounds is not coloured

- (a) Na_2CuCl_4
- (b) Na_2CdCl_4
- (c) $\text{K}_4\text{Fe}(\text{CN})_6$
- (d) $\text{K}_3\text{Fe}(\text{CN})_6$

Answer: B — (b) Na_2CdCl_4

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in Na_2CdCl_4 has the characteristic oxidation state/ligand field that produces the observed colour.

86. Acidified KMnO_4 is decolourized by

- (a) Br_2
- (b) O_3
- (c) HCl
- (d) HBr

Answer: A — (a) Br_2

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in Br_2 has the characteristic oxidation state/ligand field that produces the observed colour.

87. Which of the following compound expected to be colourless

- (a) ScO
- (b) V_2O_3
- (c) CuCN
- (d) $\text{Cr}_2(\text{SO}_4)_3$

Answer: C — (c) CuCN

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in CuCN has the characteristic oxidation state/ligand field that produces the observed colour.

88. Crystals of which pair are isomorphous

- (a) ZnSO_4 , SnSO_4
- (b) MgSO_4 , CaSO_4
- (c) ZnSO_4 , MgSO_4
- (d) PbSO_4 , NiSO_4

Answer: C — (c) ZnSO_4 , MgSO_4

Chemical reason: Isomorphous substances crystallize in the same structural type because their ions have similar charge and size. The composition represented by ZnSO_4 , MgSO_4 satisfies that structural relationship.



89. On heating $\text{Mn}(\text{OH})_2$ with PbO_2 and conc. HNO_3 pink colour is obtained due to the formation of

- (a) KMnO_4
- (b) HMnO_4
- (c) $\text{Pb}(\text{MnO}_4)_2$
- (d) PbMnO_4

Answer: B — (b) HMnO_4

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in HMnO_4 has the characteristic oxidation state/ligand field that produces the observed colour.

90. Which of the following is used as white pigment

- (a) TiO_2
- (b) V_2O_5
- (c) CuO
- (d) HgO

Answer: A — (a) TiO_2

Chemical reason: Titanium dioxide and zinc oxide are common white pigments because they have high refractive index and absorb little visible light; the option identifies the relevant white pigment.

91. Which metal oxide is used to make blue glass

- (a) Fe_2O_3
- (b) CoO
- (c) Cu_2O
- (d) NiO

Answer: B — (b) CoO

Chemical reason: Cobalt oxide compounds are used to produce blue glass because cobalt ions give an intense blue colour in silicate matrices.

92. The chemical formula for certain pyrophosphate is $\text{Ca}_2\text{P}_2\text{O}_7$, the formula of its ferric pyrophosphate will be

- (a) $\text{Fe}_2(\text{P}_2\text{O}_7)_3$
- (b) $\text{Fe}_4(\text{P}_4\text{O}_{14})$
- (c) $\text{Fe}_4(\text{P}_2\text{O}_7)_3$
- (d) Fe_3PO_4

Answer: C — (c) $\text{Fe}_4(\text{P}_2\text{O}_7)_3$

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to $\text{Fe}_4(\text{P}_2\text{O}_7)_3$. The choice is fixed by the known composition of that compound/mineral.

93. Which of the following compounds does not dissolve in ammonium hydroxide solution

- (a) AgF
- (b) AgBr
- (c) AgCl
- (d) AgI

Answer: D — (d) AgI

Chemical reason: Silver halides differ in their ability to dissolve in ammonia because complex



formation with Ag^+ must overcome their lattice energy. AgCl dissolves readily, AgBr only in concentrated NH_3 , while AgI is essentially insoluble.

94. Which of the following is non-stoichiometric

- (a) Fe_3O_4
- (b) Fe_2O_3
- (c) FeO
- (d) All the above

Answer: C — (c) FeO

Chemical reason: Non-stoichiometric compounds have variable composition caused by lattice defects, especially common among transition-metal oxides where more than one oxidation state is accessible.

95. Ferrosilicon is used in steel industry as

- (a) A flux
- (b) Scavenger of hydrogen
- (c) A reducing agent
- (d) A cutting tool
- (e) Alloying agent

Answer: E — (e) Alloying agent

Chemical reason: Ferrosilicon acts mainly as a deoxidizer/reducing additive in steelmaking: silicon has a strong affinity for oxygen and helps remove dissolved oxygen as silica/silicates.

96. In the reaction, $2\text{KMnO}_4 + 16\text{HCl} \rightarrow$

$5\text{Cl}_2 + 2\text{MnCl}_2 + 2\text{KCl} + 8\text{H}_2\text{O}$ the reduction product is

- (a) Cl_2
- (b) MnCl_2
- (c) H_2O
- (d) KCl

Answer: B — (b) MnCl_2

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives MnCl_2 . The choice is consistent with the expected oxidation state and stability of the product.

97. Which one of the following statements is correct

- (a) Manganese salts give violet borax bead test in the reducing flame
- (b) From a mixed precipitate of AgCl and AgI ammonia solution dissolves only AgCl
- (c) Ferric ions give a deep green precipitate on adding potassium ferrocyanide solution
- (d) On boiling a solution having K^+ , Ca^{2+} and HCO_3^- ions we get a precipitate of $\text{K}_2\text{Ca}(\text{CO}_3)_2$

Answer: B — (b) From a mixed precipitate of AgCl and AgI ammonia solution dissolves only AgCl

Chemical reason: The correct choice is From a mixed precipitate of AgCl and AgI ammonia solution dissolves only AgCl . It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.



98. Collin's reagent is

- (a) MnO_2/HCl
- (b) $\text{MnO}_4/\text{C}_5\text{H}_5\text{N}$
- (c) $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$
- (d) $\text{Cr}_2\text{O}_3/\text{C}_5\text{H}_5\text{N}$

Answer: D — (d) $\text{Cr}_2\text{O}_3/\text{C}_5\text{H}_5\text{N}$

Chemical reason: Collins reagent is CrO_3 complexed with pyridine in a non-aqueous solvent. It oxidizes primary alcohols to aldehydes and secondary alcohols to ketones under relatively mild conditions.

99. Which compound has coloured aqueous solution

- (a) $\text{Zn}(\text{NO}_3)_2$
- (b) LiNO_3
- (c) $\text{Co}(\text{NO}_3)_2$
- (d) $\text{Ba}(\text{NO}_3)_2$

Answer: C — (c) $\text{Co}(\text{NO}_3)_2$

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in $\text{Co}(\text{NO}_3)_2$ has the characteristic oxidation state/ligand field that produces the observed colour.

100. Zinc oxide when treated with NaOH solution gives

- (a) $\text{Zn}(\text{OH})_2$
- (b) ZnCl_2
- (c) Na_2ZnO_2
- (d) $\text{Zn}(\text{OH})_3$

Answer: C — (c) Na_2ZnO_2

Chemical reason: Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result Na_2ZnO_2 is consistent with that behaviour.

